← Back **Advanced Deep Learning Techniques - Assessment**
Graded Assignment • 15 min

Due Mar 5, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 40%. We keep your highest score.

Next item →

1 / 1 point

1. When loading the VGG16 model, what does setting include_top=False signify?

- (●) It excludes the final fully connected classification layers of the model
- ( ) It excludes the convolutional base of the model
- ( ) It loads the model without pre-trained weights
- ( ) It disables transfer learning

Correct
This is Correct. include_top=False removes the original classification layers, allowing you to add your own custom layers for the new task. For more information, refer to the video "Transfer Learning-Demo".

1 / 1 point

2. What is the purpose of freezing layers in the VGG16 model during transfer learning?

- (●) To prevent the pre-trained weights from being updated during training
- ( ) To slightly speed up the training process
- ( ) To ensure the model learns only from the new data

← Back **Advanced Deep Learning Techniques - Assessment**
Graded Assignment • 15 min

Due Mar 5, 11:59 PM IST

2. What is the purpose of freezing layers in the VGG16 model during transfer learning?

- (●) To prevent the pre-trained weights from being updated during training
- ( ) To slightly speed up the training process
- ( ) To ensure the model learns only from the new data
- ( ) To reduce the model's complexity

Correct
This is Correct. The most crucial reason for freezing layers is to preserve the pre-trained features learned from the large ImageNet dataset. For more information, refer to the video "Transfer Learning-Demo".

1 / 1 point

3. Which activation function maps the input to a range between 0 and 1 and is historically popular but suffers from vanishing gradients?

- (●) Sigmoid
- ( ) Hyperbolic Tangent (tanh)
- ( ) Rectified Linear Unit (ReLU)
- ( ) Linear

Correct
This is Correct. The sigmoid function has an S-shaped curve that outputs values between 0 and 1. It was widely used in early neural networks, but its gradients tend to become very small as the input values move away from zero, hindering learning in deep networks. For more information, refer to the

1 / 1 point

4. Which activation function is similar to sigmoid but maps the input to a range between -1 and 1?

- ○ A: Sigmoid
- ◉ Hyperbolic Tangent (tanh)
- ○ Rectified Linear Unit (ReLU)
- ○ Linear

Correct
This is Correct. The tanh function is similar in shape to the sigmoid but its output range is between -1 and 1. This centering often leads to faster convergence during training compared to sigmoid For more information, refer to the video “Activation Functions - Demo”.

1 / 1 point

5. Which of the following is a common approach in transfer learning?

- ○ Training a model from scratch with random weights
- ○ Completely discarding pre-trained models in every training iteration
- ○ Avoiding the use of neural networks
- ◉ Using a pre-trained model as a feature extractor and fine-tuning only specific layers

Correct
Correct. This is a common approach in transfer learning where the lower layers of a pre-trained model are retained, and only the higher layers are fine-tuned for a specific task. For more information, refer to the video on “Transfer Learning Techniques”.